

# Substrate Hall-algebra engine v0.4 — the negative part (antimatter) via the Drinfeld double; engine complete (4/4); + honest cyclotomic cross-check status

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## Engine v0.4 — negative part (antimatter); engine structure complete

### 0. Completing the engine

Engine lane: step 1 (coproduct = fusion/decay vertices) → step 2 (canonical basis = elementary particles + ladder) → step 3 (R-matrix = scattering) → step 4 (negative part = antimatter). This completes the ENGINE STRUCTURE — the four pieces that make the substrate Hall algebra a full process-engine.

### 1. The negative part (rigorous, standard)

The full quantum group has the triangular decomposition:

$$U_q(B_2) = U_q^+ \otimes U_q^0 \otimes U_q^-$$

- $U_q^+$  ( $E_i$ , the Hall algebra) = PARTICLES (modules/representations — what steps 1-3 built).
- $U_q^-$  ( $F_i$ , the dual Hall algebra) = ANTIPARTICLES.
- $U_q^0$  ( $K_i$ , the Cartan) = the CONSERVED CHARGES (the grading — the quantum numbers).

The Drinfeld double construction builds  $U_q(B_2)$  from the Hall algebra  $U_q^+$  and its dual  $U_q^-$ , with the Hopf pairing  $\langle E_i, F_j \rangle = \delta_{ij} / (q_i - q_i^{-1})$  ( $q = 2$  substrate-natural). So the negative part is not new input — it is the canonical dual of the Hall algebra we already have.

### 2. The physical dictionary (engine step 4)

| Structure                                           | Physical role                                             |
|-----------------------------------------------------|-----------------------------------------------------------|
| $U_q^+$ ( $E_i$ ) — positive part                   | particles                                                 |
| $U_q^-$ ( $F_i$ ) — negative part                   | antiparticles                                             |
| $U_q^0$ ( $K_i$ ) — Cartan grading                  | conserved charges / quantum numbers                       |
| bar involution / antipode ( $E \leftrightarrow F$ ) | CPT (the substrate's particle↔antiparticle conjugation)   |
| Hopf pairing $\langle E_i, F_j \rangle$             | particle-antiparticle annihilation amplitude              |
| $E \leftrightarrow F$ sector asymmetry              | baryogenesis (matter-antimatter asymmetry) algebraic home |

Antimatter is now in the engine: it is the negative part of the Drinfeld double, the canonical dual of the particle (Hall) algebra. CPT is the bar/antipode symmetry relating E and F.

### 3. Baryogenesis gets an algebraic home

Matter-antimatter asymmetry (baryogenesis) requires CP violation + the Sakharov conditions. In the engine:  
- The  $E \leftrightarrow F$  duality is exact at the symmetric point (matter = antimatter). - Baryogenesis = a substrate-level ASYMMETRY between the E (particle) and F (antiparticle) sectors — a CP-violating difference in the structure constants or the pairing. - This connects to the CKM/PMNS CP phases (we have the mixing angles; the CP phase is where the E/F asymmetry would live).

Honest: this LOCATES baryogenesis (it's the E/F sector asymmetry) but does not yet DERIVE the asymmetry magnitude — that needs the explicit CP structure (Phase 2+). The algebraic home is the deliverable; the magnitude is downstream.

### 4. Engine structure complete (4/4) — what's done, what remains

ENGINE STRUCTURE COMPLETE (the four pieces): 1. Multiplication + Green coproduct = fusion/decay vertices [?] 2. Canonical/crystal basis = elementary particles + state ladder (Lusztig positivity = they count) [?] 3. R-matrix = scattering S-matrix + braiding statistics [?] 4. Negative part = antiparticles + CPT + baryogenesis home [?]

This is Goal 1's STRUCTURE: the substrate Hall algebra is a full quasi-triangular Hopf algebra (Drinfeld double) — particles, antiparticles, vertices, scattering, conservation, statistics, CPT — all in one object.

What remains (explicit + physical): - The EXPLICIT structure constants / R-matrix entries / canonical basis at all weights (Elie's reaction table #405 + multi-week computation). - The affine extension throughout (the generation tower — tubes/imaginary roots). - The Phase-2  $A_{\text{sub}} \leftrightarrow H(Q_{B2})$  dictionary — the gate that maps the algebra's structures onto the PHYSICAL 5-tuple coordinates (region/ $\sigma_{BF}$ /chirality/charge). Until this lands, the physical IDENTIFICATIONS (which E = which particle) are the bet.

### 5. Honest cyclotomic cross-check status (per Keeper — kept alive)

Keeper: keep the cyclotomic generation route alive (it carries generations if the tube number pins  $\neq 3$ ). Honest status of the cross-check:

- $|\Phi^+(B_2)| = 4$  is a FORCED root-count (no theorem-scope issue, unlike the tubes). The chain length is also  $4 \rightarrow$  the COUNT matches, and the count is forced.
- BUT the chain VALUES  $\{\text{rank}, N_c, n_{C,g}\} = \{2, 3, 5, 7\}$  = first 4 primes are INTEGERS, not the root vectors  $\{\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2\}$ . So the value-correspondence is a SEPARATE (Mersenne/cyclotomic) fact, NOT a bijection chain  $\leftrightarrow$  roots.
- Honest conclusion: the cyclotomic route's robust forced ingredient is the COUNT ( $|\Phi^+|=4$ ); generations =  $|\Phi^+|-1 = 3$  via base/seed (Gate 1). This is cleaner than the tube route ( $|\Phi^+|=4$  is forced; the tube number is not) — BUT the full forcing still needs the structural link “why the substrate commitment chain has exactly  $|\Phi^+(B_2)|$  levels.” The count match is forced; the mechanism link is the remaining open piece.

So generation-forcing remains OPEN, with the cyclotomic  $|\Phi^+|-1$  route the better-footed candidate (forced count) — and the tube route the rep-theoretic test (Dlab-Ringel pin) that could either reinforce or break. Neither is closed; both matched. No “generations forced” headline (per #407 v0.2).

### 6. Honest scope + tier

RIGOROUS (standard): triangular decomposition  $U_q(B_2) = U_{q^+} \otimes U_{q^0} \otimes U_{q^-}$ ; Drinfeld double; Hopf pairing; bar/antipode;  $|\Phi^+(B_2)|=4$ .

FRAMEWORK (the bet): F = antiparticles,  $E \leftrightarrow F$  asymmetry = baryogenesis, Cartan grading = physical conserved charges — all ride on the Phase-2  $A_{\text{sub}} \leftrightarrow H$  dictionary for the physical identification.

ENGINE STRUCTURE COMPLETE; EXPLICIT + PHYSICAL pending: the four-piece structure is in place (Goal 1 structurally); the explicit entries (Elie) + the physical 5-tuple dictionary (Phase 2) are the remaining work to make the dynamics (Goal 2) concrete.

Cal #27 / honesty: “engine structure complete” is a STRUCTURE claim (the four algebraic pieces exist), NOT “the SM dynamics are derived.” The physical identifications are the bet, earned via the dictionary + E0/reaction-table + the generation-forcing resolution. I’m stating the structure is complete and the physics is the bet — not conflating them.

Next (my lane): Phase 2 — the  $A_{\text{sub}} \leftrightarrow H(Q_{B2})$  dictionary (the gate that converts the physical 5-tuple cells to derived; the single highest-leverage remaining item — it’s what turns “engine structure” into “physical model”). Plus support Elie’s reaction table (#405) with the engine structure.

— Lyra, Engine v0.4 negative part / antimatter filed — ENGINE STRUCTURE COMPLETE (4/4). Negative part (F) = antiparticles; full  $U_q(B_2) = \text{particles} + \text{antiparticles} + \text{conserved charges} + \text{CPT (bar/antipode)}$ ; baryogenesis =  $E \leftrightarrow F$  asymmetry (algebraic home). Goal 1 structurally complete: the substrate Hall algebra is a full quasi-triangular Hopf algebra with vertices/ladder/scattering/statistics/CPT in one object. Cyclotomic cross-check (honest):  $|\Phi^+(B_2)|=4$  forced (count), value-correspondence separate; generation-forcing still OPEN, cyclotomic  $|\Phi^+|-1$  better-footed than tubes. Remaining: explicit entries (Elie) + Phase-2  $A_{\text{sub}} \leftrightarrow H$  dictionary (the physical-identification gate). Next: the dictionary.